

A cross-cultural analysis of toolmaking skill acquisition in non-industrial societies

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1 Introduction

Toolmaking, together with bipedalism and language, was once regarded as a defining feature that makes humans distinct from other species (Oakley, 1949). It has also been constantly portrayed as the pinnacle in the evolution of cognition and creativity (e.g., Bergson, 1998; Vaesen, 2012). However, this simple notion of human uniqueness and its associated linear evolutionary narrative was overturned in the past decades by mounting evidence of toolmaking by non-human primates and corvids (Shew, 2017; Shumaker et al., 2024), particularly in the domain of food foraging. As pointed out sharply by Shumaker et al. (2024: 246-248), even some of the higher-order cases of tool manufacture, such as using existing tools to make other tools, using tools in combinations for a single task, and cooperation in making tools, have been demonstrated to be not uniquely human in an absence/presence manner. From a comparative perspective, it seems that the recruitment of toolmaking process for novel functions (Davidson & McGrew, 2005) has not yet been recorded in any species other than the human lineage. In terms of time allocation of tool-use and toolmaking as well as the level of dependency in caloric intake (Shipman, 2010), toolmaking occupies a central niche in our daily activities as compared with other toolmaking species.

On top of the basic survival need, it is also critical to human experience in the sense that a bio-cultural feedback loop is formed through toolmaking that shapes our minds and bodies in turn (Birch, 2021; Malafouris, 2012, 2021; Stout, 2021), as demonstrated by an extensive corpus of literature revolving around stone tools due to their longevity and great preservation. For example, a series of neuroimaging studies designed by Stout and colleagues (Stout & Chaminade, 2007; Stout & Hecht, 2023; Stout & Khreisheh, 2015) revealed the shared neuroanatomic foundations and differential cognitive demands of various lithic technologies, highlighting the evolutionary importance of perceptual-motor adaptation in technological evolution. Stone toolmaking and use has also been identified as a major selection force driving the diachronic changes in human hand morphology, including both bone and soft tissue anatomy, based on multiple biomechanical/kinematic studies in the context of experimental archaeology (Dunmore et al., 2023; Karakostis et al., 2021; Kivell et al., 2023; Marzke, 2013)

Yet the skill of toolmaking is not born with us — naive individuals need to acquire a given skill either through observing and learning from others (social learning) or repeated trials and errors (individual learning). Very often the skill attainment, particularly this type of motor skill involv-

ing extensive hand-eye coordination, requires a combination of these two learning strategies. Although this behavior is of great significance in human evolutionary studies as argued above (Buckley, 2023; Derex & Morgan, 2023), the mass machinery manufacturing process of tools introduced by industrialization weaken the necessity of handicraft practices and their inter-generational transmission in our contemporary daily life. Consequently, ethnographic narratives depicting the subsistence and economy of non-industrial societies, mostly written during the nineteenth and twentieth centuries, provide us a precious opportunity to investigate this topic, among which hunter-gatherer societies represent the most well-studied and well-published sample.

Embedded within the century-old anthropological research tradition revolving around childhood, the learning dynamics of hunter-gatherers have garnered considerable attention in the past two decades due to the burgeoning literature of cultural evolutionary theories (Hewlett et al., 2024). Garfield et al. (Garfield et al., 2016) conducted the first systematic investigation on hunter-gather social learning, involving 982 ethnographic texts from 23 different communities. In this research, they characterized the commonalities of learning behaviors across diverse domains regarding their transmission mode, forms of teaching, and learners' age. Within the acquisition of subsistence skills, which dominated the early and middle childhood, they concluded that oblique transmission becomes more important when compared with vertical transmission with the increasing age of learners. Subsequently, Lew-Levy and her colleagues narrowed down the learner' identities to children and thoroughly examined their learning patterns of subsistence skill, of which toolmaking skill is a subset, depicting a more detailed picture regarding the dominant learning processes and skill types across each developmental periods (Lew-Levy et al., 2017; Lew-Levy, Milks, et al., 2020).

Drawing upon both the cross-cultural analysis of existing ethnographic texts and new systematic field data collected from certain foraging communities, this children-centered research line of inquiry (Lancy, 2017) remained to be the most fruitful one in the past few years, emphasizing the evolutionary significance of object/toy as a medium of learning (Lew-Levy et al., 2022; Riede et al., 2023) and the often ignored strategy of peer-learning (Lew-Levy, Kissler, et al., 2020; Lew-Levy et al., 2023; Lew-Levy & Amir, 2024). In addition, other researchers conducted either literature survey (Bira & Hewlett, 2023) or in-depth systematic observation (Schniter et al., 2023) on the knowledge transmission patterns of societies featuring other forms of subsistence

strategies. For example, Bira and Hewlett (2023) compared the cultural learning patterns between hunter-gatherers and pastoralist children and found that peer-learning was less likely to be documented in the latter compared with the former context. Based on long-term field data of Tsimane forager-farmers, Schniter et al. (2023) identified the importance of older same-sex relatives and multigenerational pedagogy in the successful reproduction of crucial skills like handicrafts.

Some pressing issues still remain unsolved despite previous intellectual efforts on the topic tool-making skill acquisition. First and foremost, the extremely low contribution of individual learning in subsistence skills, particularly toolmaking, which according to Garfield et al. (2016) only occupy 3.7% of all cases. There could be several reasons behind it. An obvious one is that ethnographers may systematically care more about communal activities and it's challenging to follow and observe individuals. The other source of bias could be coding. And that what I will try to improve in this project. I have a coding scheme that provide a more detailed evaluation of the learning processes. Second, the unbalanced representation of hunter-gather societies versus all other subsistence types within non-industrial societies mask our understanding because tool-making (by hand) is not unique to hunter-gatherers.

To understand the diversity and regularity in the process of human toolmaking skill acquisition, this project analyzed 170 non-industrial societies displaying varying subsistence strategies around the world using the eHRAF World Cultures database, with a particular focus on developmental contexts and transmission biases involved in the learning processes of toolmaking skills. This study is exploratory and descriptive in nature, focusing on systematically summarizing and presenting the available data without attempting to infer causal relationships or test predictive models. Descriptive studies are essential for establishing a foundational understanding of complex datasets, identifying patterns, and guiding future research directions by highlighting key trends and data limitations. (Gerring, 2012).

This study investigates the processes of toolmaking skill acquisition in non-industrial societies, focusing on fundamental aspects of learning: who, where, when, how, what, and why. For the “who” aspect, the author examined the transmission modes and concrete social relationships between the learner and the model. The “where” question addresses the spatial context of learning, such as whether it occurs indoors or outdoors. The “when” explores the ages of both the learner and the model. For the “how,” the author investigated the degree of social interaction

involved in the learning process. The “what” focuses on the content of transmission, such as pottery-making, woodworking, or other craft activities. Lastly, the “why” examines the motivations behind a learner’s choice of model—whether it is based on the model’s recognized expertise, convenience, or familial ties.

In addition to addressing the six fundamental research questions, this project also undertakes a missing data pattern analysis to evaluate the completeness and reliability of the dataset. Furthermore, a qualitative analysis of the ethnographic texts was conducted to provide contextual depth and enrich the interpretation of the database. Together, these approaches offer a more comprehensive understanding of the data, enhancing both the analytical rigor and the interpretive framework of this study.

2 Methods and materials

2.1 Searching strategy

This study analyzed a global sample composed of 170 non-industrial societies indexed in the eHRAF World Cultures database, which include 45 hunter-gathers, 36 horticulturalists, 29 other subsistence combined, 28 intensive agriculturists, 14 primarily hunter-gatherers, 13 agropastoralists, and 5 pastoralists cultures (**Figure 1**). Societies labeled as commercial economy are pre-excluded here since they are often associated with industrialization to some degree. Because the eHRAF database is organized using the Outline of Cultural Materials (OCM) coding scheme, the searching strategy focuses exclusively on one OCM code that is most relevant to the research question, namely OCM 868 “Transmission of Skills”. Under this OCM code, this project uses the following keywords for searching, including tool, make/made/making, produce/producing, manufacture/manufacturing, create/creating, construct/constructing, assemble/assembling, fabricate/fabricating, shape/shaping, forge/forging, form/forming. These keywords together yields 3060 paragraphs, among which approximately 30-40% are repetitive.

Focusing on the toolmaking skill acquisition in non-industrial societies, this project adopts the following exclusion criteria when screening the paragraphs yielded in initial search, including archaeological and historical literature sources, cases of purely tool use, instance where local people taught the ethnographer how to make tools, instances where it was clearly stated a new

form of government-initiated school to save a certain craft, instances where individuals are taught how to make Christianity-related artifacts such as altar, and instances where indigenous people mentioned that they learn a specific craft from museum pieces or photographs. However, these cases may appear as contextual information in the appendix to provide more details of the technologies themselves. In the end, the eligible paragraphs and their associated contextual paragraphs were compiled as an 290-page and geographically organized appendix available as supplementary materials of this article.

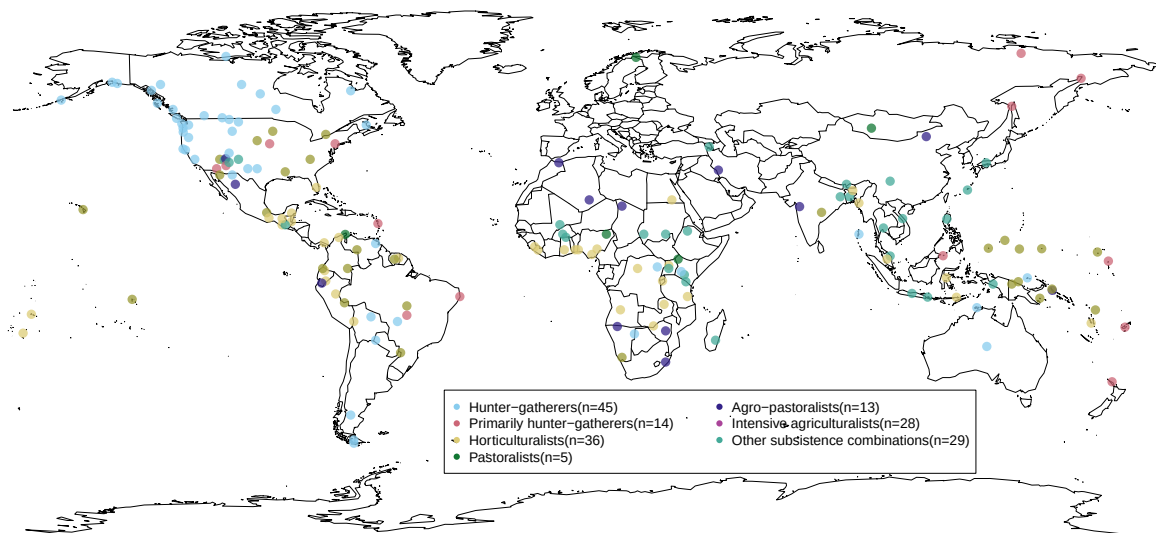


Figure 1: The spatial distribution of non-industrial societies covered in this study (n=170). It should be noted that these coordinate data represent language ‘centroids’ from the open-access language documentation site glottolog.org instead of the real geographical locations of given communities. These language centroids essentially represent the center point of the area where an relevant language is spoken.

2.2 Coding scheme

In the ethnographic coding, the acquisition of a specific toolmaking skill (e.g., clothes making) documented in one culture is regarded as an instance (i.e., a single row in the spreadsheet). When multiple sources by either same or different authors all mentioned relevant information regarding how certain technology is transmitted in a society, the coding scheme treated all of

them as “true” information, assuming that it is possible for a technology to have multiple channels of transmission unless explicitly stated otherwise. For example, source A claim that pottery making in society X is transmitted between parents and offspring, while source B claim that pottery making in the same society is transmitted between grandparents and grandchildren. In this case, the transmission mode of this observation/row should be coded as “mixed” instead of “vertical” or “oblique”. The only exception is age group of learner, for which the lowest age group is adopted (see the full coding scheme below for more details).

To ensure the reliability of coding results, the whole appendix was coded twice by the author until all discrepancies were resolved. The author employed a dual-system recording approach with some key variables such as technology, location, age of learners and models, learning process, etc. Specifically, the author first documented the descriptions provided by ethnographers and then categorized the information in a separate column for quantitative analysis. It should be noted that “NA” is used in all variables when there is not enough information to make the relevant inference, which is different from absence of the trait of interest. The following is a complete description of the coding scheme used in this project.

1. **Area:** This variable follows eHRAF’s classification system.
2. **Society:** This variable follows eHRAF’s classification system.
3. **OWC Code:** The Outline of World Cultures (OWC) is a classification system of the cultures of the world, which is used by eHRAF to organize its databases. It is used here as a unique identifier to merge the geographic coordinate data from D-PLACE for the purpose of mapping.
4. **Subsistence:** This variable follows eHRAF’s classification system, including hunter-gatherer, primarily hunter-gatherer, horticulturalist, pastarolist, agro-pastarolist, intensive agriculturalist, other subsistence combinations. The category of commercial economy is excluded. The definitions of these terms are available at the official eHRAF documentation here (<https://ehrafworldcultures.yale.edu/help/LearnMore/subsistence.html>).
5. **Year of observation:** The years when the given society was observed by the ethnographers. This information is readily available under the name “field date” in the meta information of eHRAF World Cultures.

6. **Location:** The location where the learning event happens according to the ethnographer's description.
7. **Location (type):** 1) indoor: learning happens inside a building; 2) outdoor: learning happens outside the buildings.
8. **Age of learner:** The learner's age according to the ethnographer's description.
9. **Age group of learner:** 1) Infancy and Early Childhood (0-5 years old), 2) Middle Childhood (6-12 years old), 3) Adolescence (13-16 years old), 4) Adulthood (>16 years old). For analytical purpose, this category does not concern the cultural differences on the perception of developmental differences. When an age range is provided, the minimal age will be used in the classification of age group.
10. **Number of learner:** The number of learners involved, which can either 1) single or 2) multiple.
11. **Age of model:** The model's age according to the ethnographer's description.
12. **Age group of model:** 1) Infancy and Early Childhood (0-5 years old), 2) Middle Childhood (6-12 years old), 3) Adolescence (13-16 years old), 4) Adulthood (>16 years old).
13. **Number of model:** The number of models involved, which can either 1) single or 2) multiple.
14. **Transmission modes:** 1) horizontal transmission: social learning from individuals of the same generation, age group, or cohort, roughly within 4–5 years of age; 2) vertical transmission: children learning from their parents; 3) oblique transmission: social learning between individuals of distinct generations or age groups typically from an older generation to a younger generation. It can happen within or beyond extended kin group, such as learning from grandparents, uncles, or other adults in the local population; 4) mixed: different transmission modes occurred together in the learning process of a given technology.
15. **Social relationship:** The relationship between the learner and the model according to the ethnographer's description. For example, mother-son/father-daughter/etc.
16. **Transmission bias:** This category follows the framework developed by Kendal et al. (2018). 1) kin-based bias: learning a cultural trait from people who are your kin members, 2) suc-

cess-based bias: learning a cultural trait from people because they are successful in this technology, 3) prestige-based bias: learning a cultural trait from people because of their high social status; 4) conformist bias: learning a specific type of knowledge from other people because it is displayed by the majority of local population; 5) mixed: different transmission biases occurred together in the learning process of a given technology.

17. **Learning process**: The learning process according to the ethnographer's description, such as "teach", "imitate", "observe", etc.
18. **Learning process type**: 1) individual learning: Individual exhibits repeated attempts to learn a skill or develop new skills or knowledge on his or her own, including trial and error and individual practice; 2) observational learning: learner directly observes the actions of models and attempts to replicate the observed actions, and no direct interaction between the model and learner is observed or mentioned; 3) interactive learning: this category features the verbal, gestural and bodily interactions between models and learners, be it formal ("teaching") or informal ("group play"). It should be noted that these three categories are not mutually exclusive. On the contrary, these three categories are inclusive of each other, where interactive learning is built on both observation and individual practice, and observational learning involves individual practice.
19. **Time spent**: The time spent on learning according to the ethnographer's description.
20. **Technology**: The subject of learning according to the ethnographer's description.
21. **Technology type**: The types of technology were determined *post hoc* from collected ethnographic texts based on their frequency, including agriculture equipment making, bag making, basket making, boat making, clothes making, fishing devices making, general tool-making, hammock making horse harnesses making, mat making, metal working, miscellaneous, musical instrument making, net making, ornament making, pottery making, raw material processing, ritual item making, rope making, stone working, toy making, trap making, vehicles making, vessel making, weapon making, weaving, woodworking. These categories are not mutually exclusive, and some categories are single out either because of cultural significance or the lack of relevant information. For example, weaving is a commonly documented skill that can be used in the manufacture of clothes, bag, basket, hammock, mat, etc, but it has its own category because no other contextual information is

available to differentiate between them. Miscellaneous is used when the frequency of a specific toolmaking skill is lower than three. General toolmaking is an umbrella terms for phrasing like “utensils” or “tools” in the original text.

22. **Raw material:** The raw materials were also determined *post hoc* from collected ethnographic texts based on their frequency, including stone, clay, plant fiber, wood, bones, leather, composite, etc.
23. **Note:** This records interesting information that is not systematically documented in other variables, such as the difficulty of a skill as perceived by local people and ethnographers, access to raw material, and the restriction on the social status involved in the learning such as gender, wealth, religious status, etc.
24. **Reference:** This provides the relevant references, which can be linked to the relevant paragraph in the appendix.

2.3 Statistical analysis and visualization

To assess the patterns of missing data, the author conducted a comprehensive missing data analysis using a combination of visualization techniques and statistical testing. A heatmap and an UpSet plot were generated with the R package **naniar** (Tierney & Cook, 2020) to visually explore the distribution and intersections of missing values across variables. The heatmap highlighted the overall distribution of missing data, while the UpSet plot provided insights into how missingness co-occurs among specific variables. Additionally, Little’s MCAR test was performed to statistically evaluate whether the missing data were missing completely at random (MCAR). To ensure transparency and reproducibility (Marwick, 2017), the dataset and the full analysis code have been uploaded to a dedicated GitHub repository (<https://github.com/Raylc/HRAF-2022>).

3 Results

3.1 Descriptive results

3.1.1 What

Figure 2A displays the frequency distribution of various tool-making skills documented in non-industrial societies. The data reveal significant variation in the representation of different craft activities. Clothes-making emerges as the most frequently recorded skill ($n = 64$), followed by basket-making ($n = 49$), weapon-making ($n = 44$), and pottery-making ($n = 38$). Weaving ($n = 28$) and woodworking ($n = 26$) are also prominently represented. In contrast, certain specialized skills, such as vehicle-making and agricultural equipment-making, appear infrequently ($n = 3$ each). This distribution suggests that daily utilitarian crafts, particularly those related to clothing and storage, are more commonly documented, possibly reflecting their widespread necessity and frequent practice. In contrast, more specialized or context-specific crafts, such as ritual item-making ($n = 6$) and horse harness-making ($n = 4$), are less commonly reported. The variation in frequency may reflect the cultural importance of certain skills and/or the observational bias focusing on more pervasive or visible activities within these societies.

If we reorganize the classification of toolmaking skills based on their raw materials involved, the results reveals a significant disparity in their distribution (**Figure 2B**). The majority of entries fall under the “NA” category ($n = 182$), indicating a substantial portion of missing data for this variable. Among the recorded materials, plant fiber ($n = 157$) and wood ($n = 63$) are the most frequently documented, suggesting their prevalent use or better preservation in the archaeological record. In contrast, materials such as clay ($n = 39$), metal ($n = 18$), leather ($n = 16$), stone ($n = 7$), composite ($n = 4$), and bone ($n = 2$) appear far less frequently. This uneven distribution could be a mixed results shaped by raw material accessibility and processing difficulty, potential cultural preferences in material use, as well as observational biases. Moreover, the prominence of clothing and basket-making technologies, alongside the frequent use of plant fiber and wood—materials prone to poor preservation—suggests that significant aspects of past material culture are underrepresented in the archaeological record. This highlights the importance of integrating ethnographic data to reconstruct technological practices that are archaeologically invisible, offering a more comprehensive understanding of past human behavior ([Desmond, 2022](#); [Gilligan](#)

et al., 2024; Langley & Suddendorf, 2020).

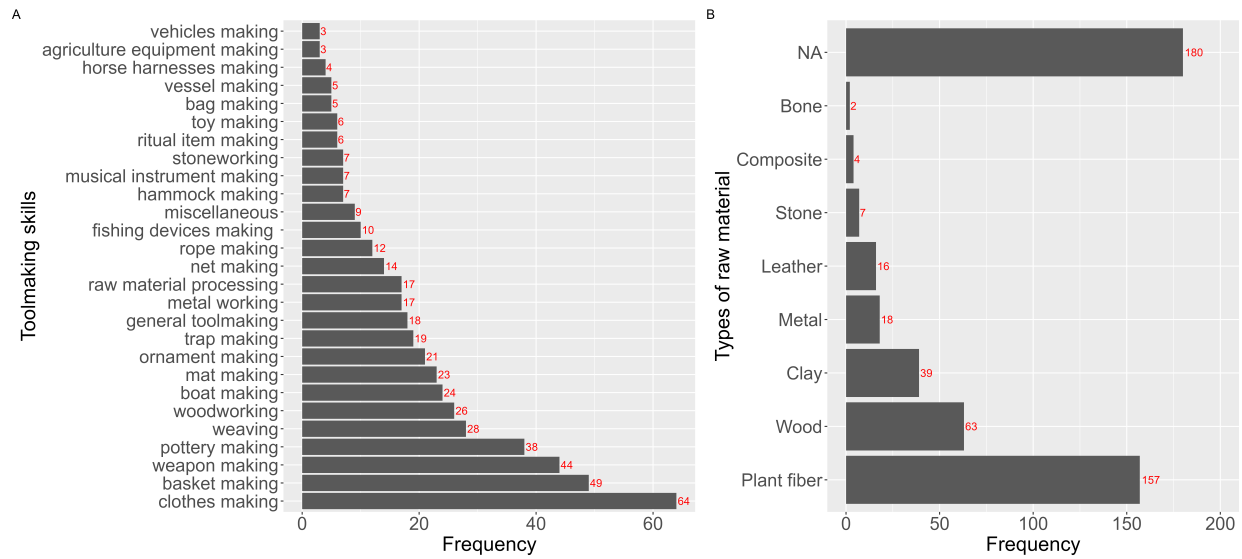


Figure 2: The frequency of toolmaking skills (A) as well as main raw materials involved (B) as documented in ethnographic texts.

3.1.2 Who

Here the author visualized the distribution of different modes of cultural transmission (**Figure 3**). The results indicate that vertical transmission, where skills are passed from parent to child, is the most frequently documented mode ($n = 162$). This is followed by vertical/oblique transmission ($n = 112$), where learning occurs both from parents and other adults within the community. Purely oblique transmission—learning from non-parental adults—accounts for 59 instances. Horizontal transmission, involving learning from peers, appears less frequently ($n = 10$), and mixed modes combining horizontal and oblique transmission are even rarer (e.g., horizontal/oblique: $n = 6$; oblique/horizontal: $n = 1$). Additionally, some cases report a combination of all modes ($n = 11$). Notably, 118 instances lack specific information on transmission mode, highlighting gaps in ethnographic reporting.

These results suggest a strong reliance on kin-based learning pathway, particularly vertical transmission, for toolmaking skills in non-industrial societies. It provides strong empirical support of Derek and Morgan's (2023) recent claim regarding the importance of vertical channel in skill transmission when compared with knowledge transmission. Taking a closer look at the social relationship between models and learners (see supplementary table 1), the author also found that the majority cases under the categories of oblique transmission involve grand-

parents, aunts, or uncles, which are also close kin members of the learner. This pattern may reflect the importance of familial and inter-generational knowledge transfer in maintaining craft traditions. Interestingly, the data show limited evidence for peer-to-peer (horizontal) transmission as contrary to what has been reported by Lew-Levy and her colleagues ([Lew-Levy, Kissler, et al., 2020](#); [Lew-Levy et al., 2023](#); [Lew-Levy & Amir, 2024](#)). It is possible that peer play activities or enculturation through interactions with older siblings falling into a similar age range are not that vital in the domain of toolmaking skills as compared with knowledge domains such as social and gender norms ([Lew-Levy et al., 2018](#)), considering the fact that several skills either require many years' practice to master ([Bock, 2002](#); [Gurven & Kaplan, 2006](#); [Kaplan et al., 2003](#)) and/or a certain degree of physical strength and motor control that is only available during or after adolescence ([Bird & Bliege Bird, 2002](#); [Bliege Bird & Bird, 2002](#); [Jones & Marlowe, 2002](#)).

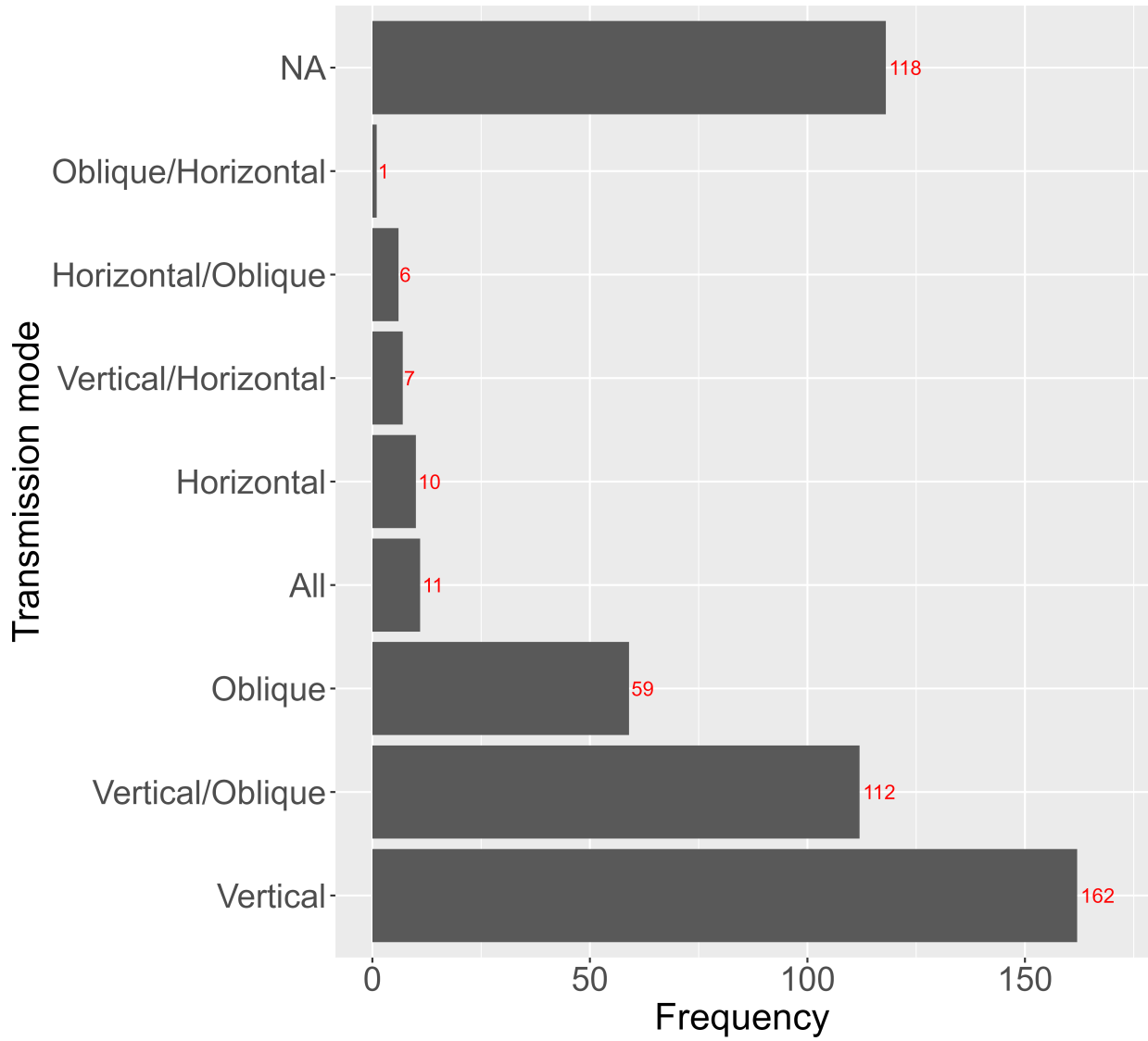


Figure 3: Transmission modes of toolmaking skills documented in ethnographic texts.

3.1.3 Where

As displayed in (Figure 4A), the majority of ethnographic cases ($n = 421$) lacked explicit information regarding the learning locations of toolmaking skills. Among the cases with documented locations, learning occurred most frequently indoors ($n = 38$), followed by outdoor settings ($n = 25$), with only two instances involving mixed indoor-outdoor contexts. Figure 4B further illustrates the relationship between learning environments and specific tool-making activities using a treemap. Indoor learning environments are predominantly associated with activities such as clothes making ($n = 6$), basket making ($n = 5$), and metal working ($n = 5$). In contrast, outdoor

learning environments are primarily linked to boat making (n = 4), trap making (n = 3), and net making (n = 2). Only two activities, metal working and stoneworking, were reported in both spatial contexts.

This pattern suggests that the physical setting of skill transmission is potentially tied to the nature of the craft. Activities involving fine, detailed work or those requiring specialized tools and materials are more often learned indoors, whereas tasks demanding larger spaces or environmental resources, such as boat or trap making, are typically learned outdoors. However, it is also possible that this distribution might simply reflect observers’ biases given the size of unreported cases.

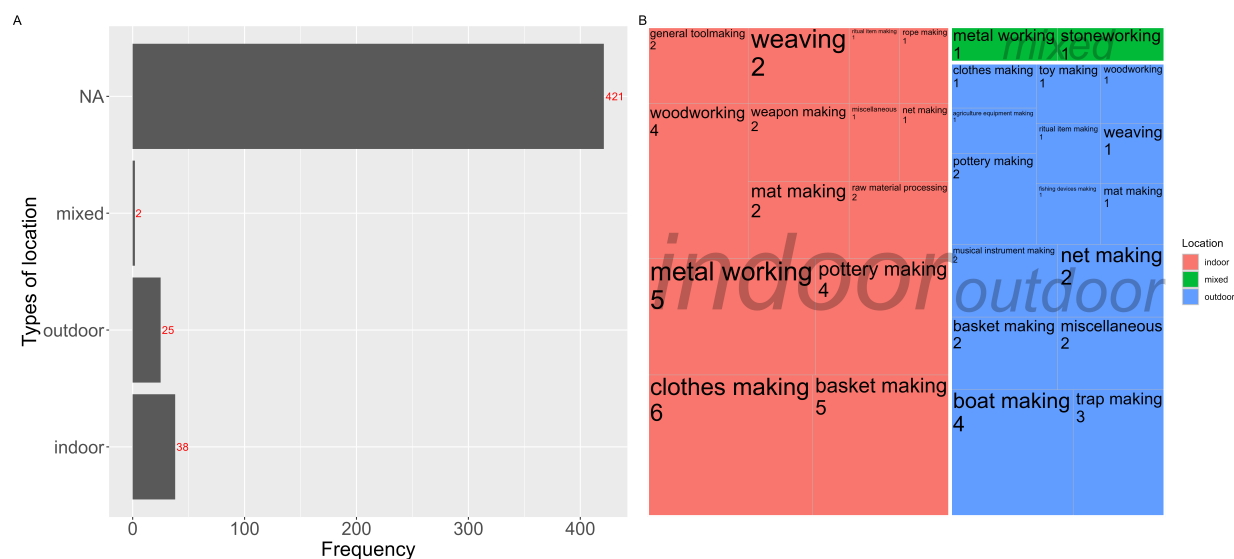


Figure 4: Location of toolmaking skill acquisition documented in ethnographic texts. A displays the frequency of learning locations. B is a treemap demonstrating the connections between craft categories and location locations.

3.1.4 When

The analysis of toolmaking skill learners’ developmental contexts (**Figure 5**) revealed that most learning instances (n=274) did not have specific age-related information documented. Among the cases where age was specified, middle childhood (ages 6-12) showed the highest frequency of learning episodes (n=128), followed by adolescence (ages 13-16) with 47 instances. Infancy and early childhood (ages 0-5) accounted for 34 documented cases, while adulthood showed the lowest frequency with only 3 recorded instances. This distribution suggests that the transmission of tool-making knowledge in non-industrial societies occurs primarily during middle childhood,

a critical period in developmental psychology featuring multiple cognitive milestones such as advanced episodic memory (Ghetti & Bunge, 2012), theory of mind (Osterhaus & Koerber, 2021), and executive function (Wilson et al., 2018). Again, the substantial number of unspecified age cases (NA) indicates potential limitations in the ethnographic documentation of age-specific learning processes.

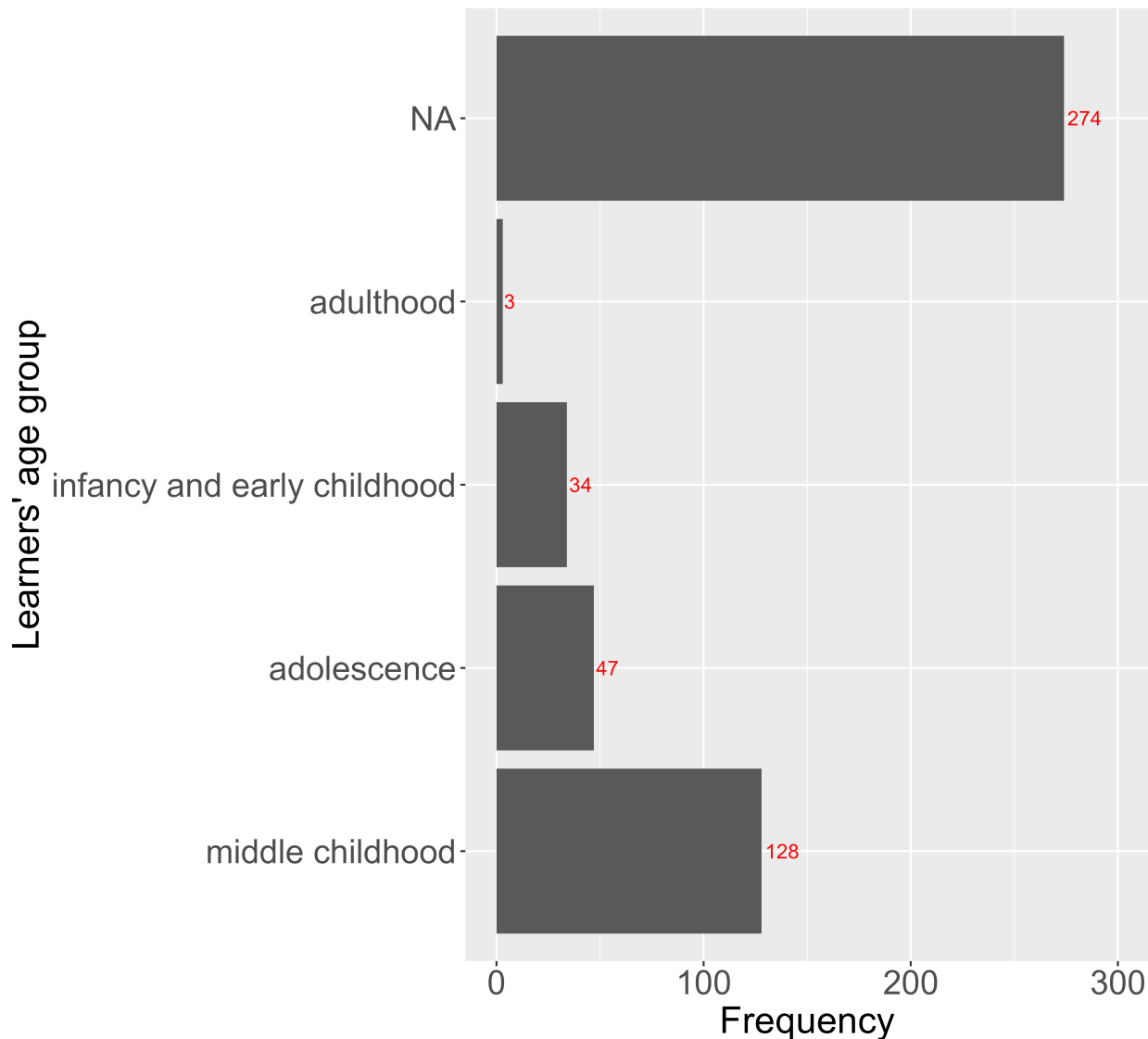


Figure 5: Developmental contexts of toolmaking skill acquisition documented in ethnographic texts.

3.1.5 How

Figure 6 presents the distribution of learning processes for toolmaking skills as . Interactive learning, which involves direct engagement between teacher and learner, was the most common

method documented with 304 instances. Observational learning, where individuals learned by watching others without direct interaction, occurred in 62 cases. Individual learning, characterized by self-directed experimentation and practice, was recorded in 16 instances. In 104 cases, the specific learning process was not identified. These results indicate that the transmission of tool-making knowledge in non-industrial societies relies heavily on social interaction, be it verbal or gestural, with observation serving as a secondary but significant learning method. The relatively low frequency of individual learning confirms the observer's bias towards social activities as opposed to following individuals as mentioned in the introduction, which by no means suggested that individual trial-and-error learning is not important in the acquisition of toolmaking skills. Rather, given the inclusive nature between three categories in this coding scheme, this result suggests that practice is integral to the learning process but pure individual practice without social interaction is rarely documented.

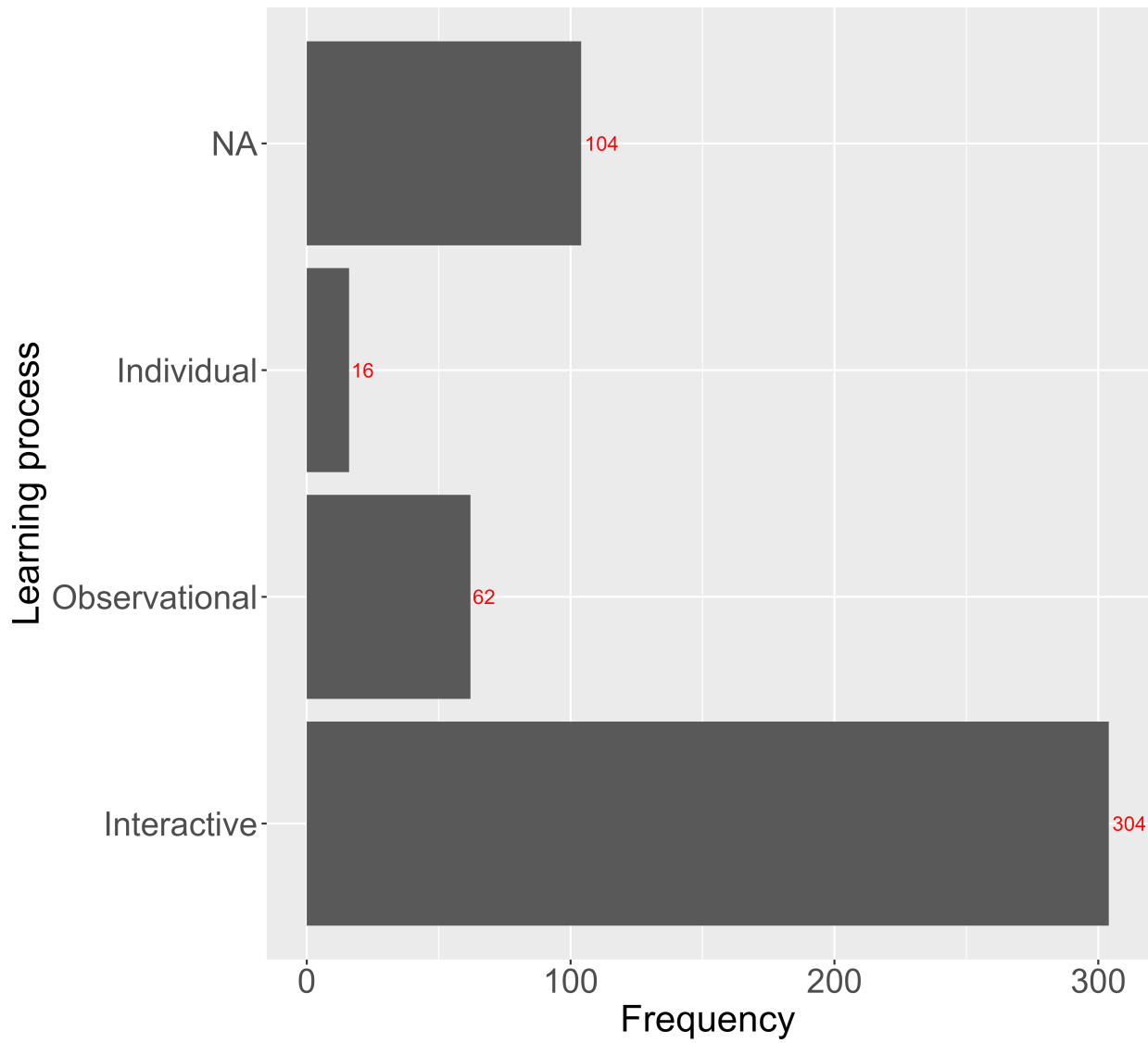


Figure 6: Learning processes of toolmaking skill acquisition documented in ethnographic texts.

3.1.6 Why

The distribution of transmission bias (**Figure 7**) agree with the result of transmission mode analysis as illustrated in the who section above. Kin-based transmission emerged as the predominant pattern with 288 documented cases, suggesting that tool-making skills are most often passed down within family units. Mixed transmission strategies, which combine different learning approaches, were observed in 33 cases, while success-based transmission occurred in 28 instances. Prestige-based transmission, where learners chose models based on their social status, was documented in 8 cases, and familiarity-based transmission appeared in only 3 instances. A

substantial number of cases ($n=126$) did not specify the transmission bias (NA). Again, this distribution shows that kinship plays a central role in the cultural transmission of tool-making skills, while other transmission biases occur less frequently. This finding is particularly interesting because traditionally the field of cultural evolutionary model focuses mainly on success/prestige bias or conformist bias (copying the majority), while kin-based bias remains understudied. A potential explanation is that success/prestige bias ([Chellappoo, 2021](#); [Chudek et al., 2012](#); [Jiménez & Mesoudi, 2019](#)) or conformist bias ([Henrich & Boyd, 1998](#); [McGuigan et al., 2012](#); [Wakano & Aoki, 2007](#)) may be more important in other cultural domains like the transmission of beliefs or ascetic preferences, while toolmaking usually is a long-term learning process that featuring more intimate social relationship.

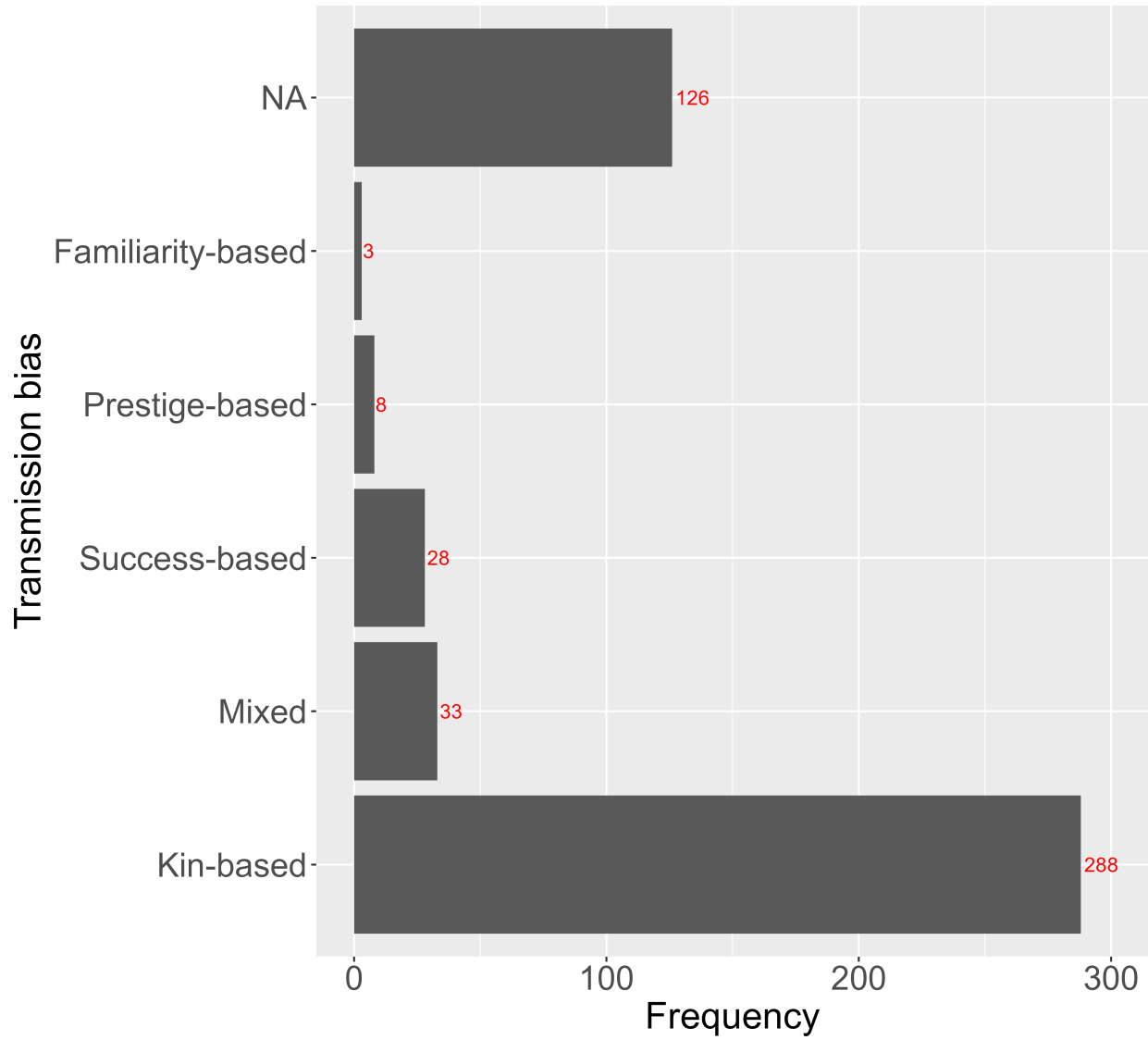


Figure 7: Transmission biases of toolmaking skill acquisition documented in ethnographic texts.

3.2 Missing pattern analysis

The missing data heatmap (**Figure 8**) visualizes the distribution of missing values across observations and variables. High proportions of missing data are evident in variables like “Time.spent” (92%), “Location.type” (87%), “Number of learner (87%)” “Age.group.of.learner” (56%), and “Number.of.model” (40%). This was further verified by the result of Little’s statistical test for missing completely at random (MCAR) data ($p < 0.001$), suggesting this dataset is not missing completely at random.

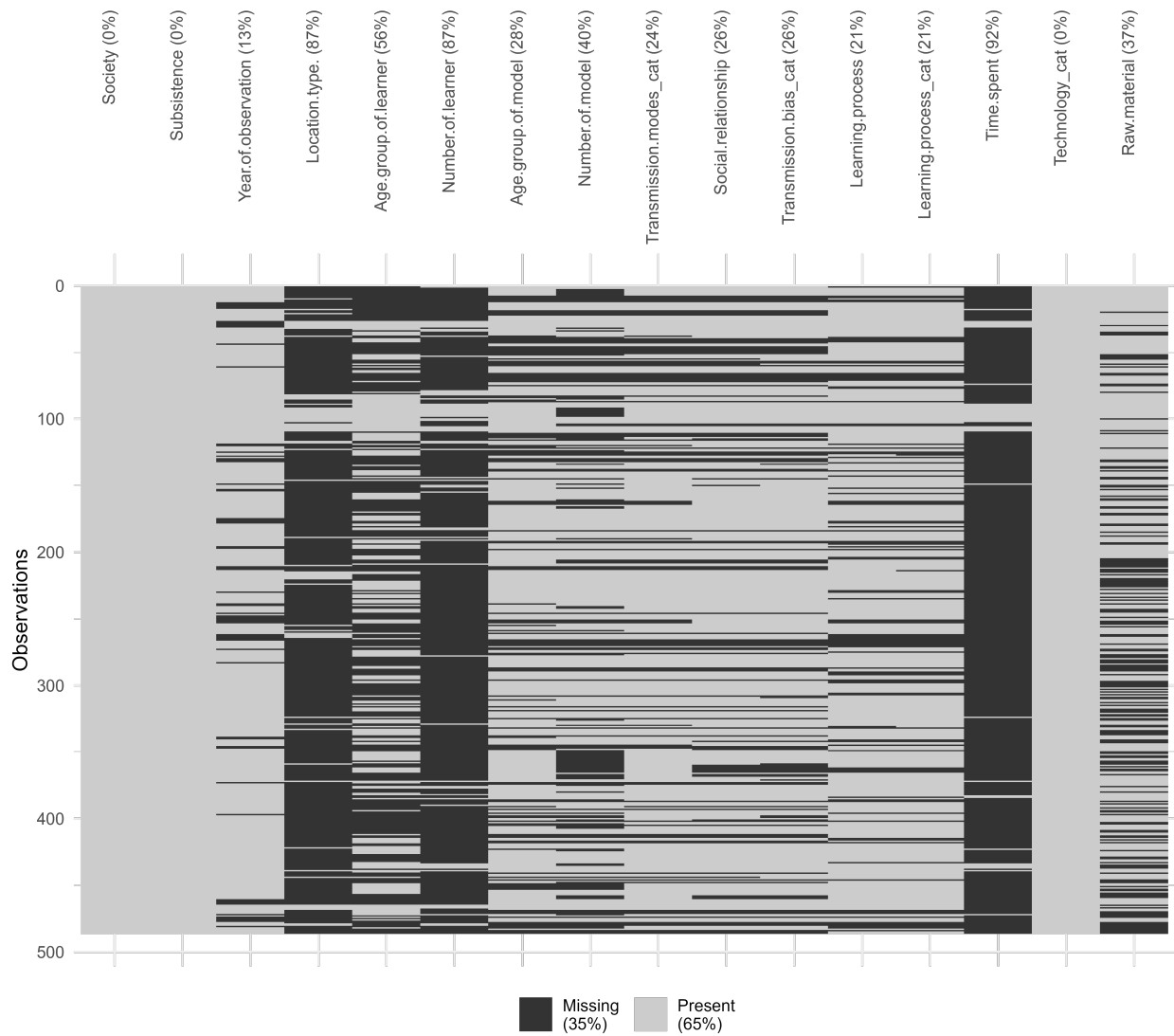


Figure 8: Missing data heatmap. Darker segments represent missing data, while lighter segments indicate available data.

The UpSet plot (**Figure 9**) further illustrates the extent and combinations of missing values among the variables. Notably, 79 observations feature missing data of the four variables with the highest percentage of missing values, including Time spent, number of learner, location type, and age group of learners. In 39 instances, it was found that nine out of ten variables with the highest percentage of missing values are co-missing, and similar patterns occur repeatedly in several intersecting groups. The clustering of missingness across specific variables suggests non-random patterns, possibly due to the varying research interests of the observers as well as the complexity of documenting certain social or learning contexts in the fieldwork.

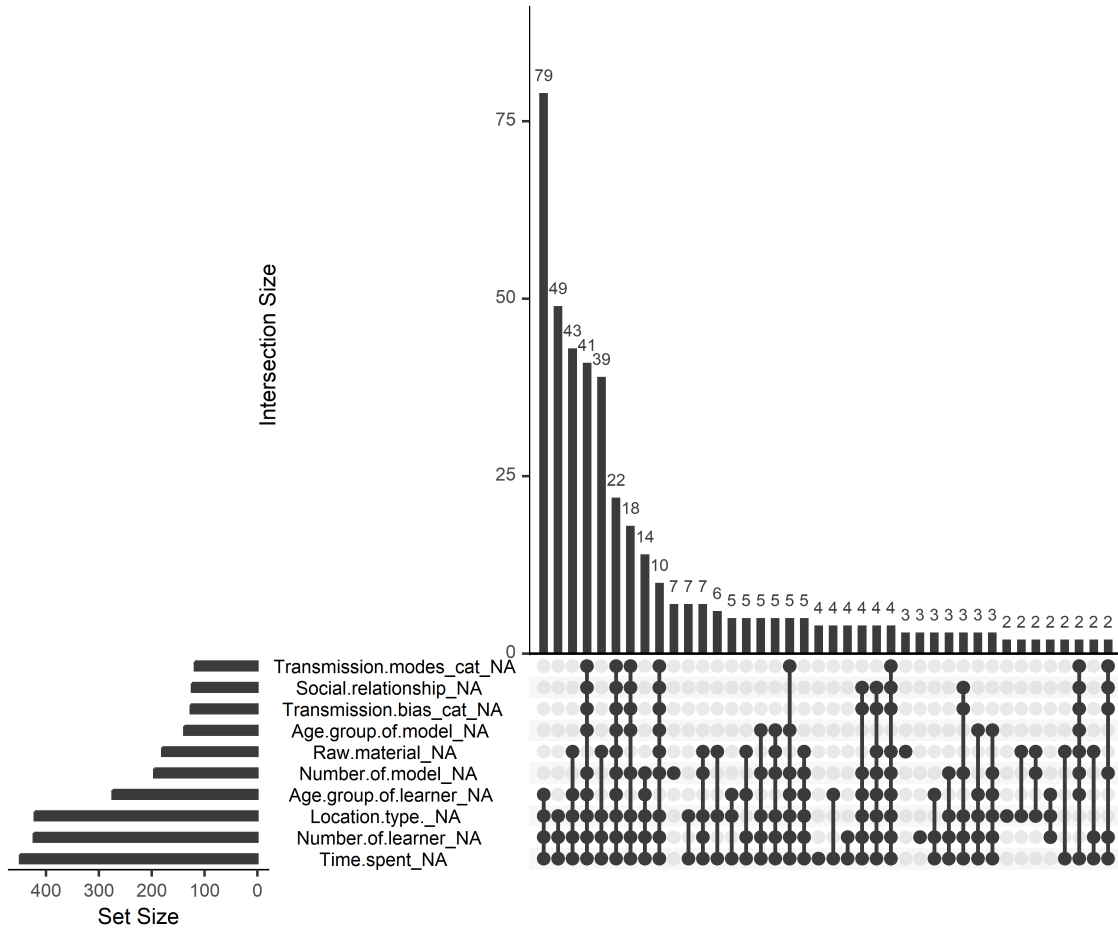


Figure 9: The UpSet plot displaying the co-occurrence of missing variables across observations. Ten variables with the highest percentage of missing values are visualized here.

4 Qualitative insights

While the quantitative results provide a broad overview of the distribution and frequency of documented technologies and raw materials, they do not fully capture the complexity of how these technologies were learned, transmitted, and practiced. During the coding of ethnographic data, unexpected patterns and nuanced insights began to emerge, revealing aspects of technological knowledge and skill that were not initially targeted by pre-designed questions. These qualitative findings offer a deeper contextual understanding that complements the quantitative analysis, highlighting the importance of flexible, data-driven interpretation in uncovering the social dynamics of technology use and production. More specifically, these qualitative data will provide complimentary insights to four of our six basic questions, including, when, where, how, and why.

4.1 Protracted and staged learning trajectories

This study partly confirms Dallos's (2021) sharp observation on the significant role of protracted learning in the formation of human culture, as several HRAF ethnographic sources suggested that the learning of toolmaking skills spans multiple developmental periods in an almost structural manner, where each periods feature different approaches to learning and varied task priorities. This applies to scenarios where people learn different different domains of technology at different ages as well as people acquiring one toolmaking skill learn various technological components in a developmentally scaffolding manner. For instance, in Lancy's (1996: 161-162) classical ethnography on Kpelle children development, he noted that:

“Children learn three types of tasks at three different ages. They begin performing such rudimentary tasks of rice farming as cooking, pounding rice, and chasing away birds at a very early age, usually before they are 9. More difficult tasks, such as hoeing on the farm, net weaving, and trap making, are begun around age 9 and mastered by age 13. Rarer, more complex skills such as cloth weaving, bag weaving, blacksmithing, and medicine making are acquired in the late teens and may be learned after marriage.”

Canoe making, a skill that demands not only the acquiring of technical know-how but also the development of proper physical strength to handle heavy logs, best illustrates the case of long training period required for a single toolmaking skill. For example, Wilbert (1976: 360) commented that among the Warao hunter-gatherers' skill training for men last until 25 years old and he further(Wilbert, 1977: 23) wrote that:

“A boy is associated with canoe-making from early childhood on, initially as spectator, when he accompanies his father and the other men of the family to the forest and watches the production process. Later, as an adolescent, he picks up the axe himself and participates in the process according to his increasing dexterity. By the time he gets married, a young man commands the basic skills and may even have a small canoe to call his own.”

This pattern of prolonged and staged learning has also recurrently occurred in other ethnographic observations on other indigenous communities such as Chuuk/Truk (Goodenough,

1951: 53-55) in Micronesia, as well as Kapauku (Pospisil, 1958: 40-41) and Trobriands (Austen, 1945: 194-195; Scoditti, 1989: 56) in Melanesia (e.g., “15-20 years of apprenticeship”). However, given the inherent constraint that most ethnographic sources used here only provided a fragmented narrative on craft learning, it is still unclear how common is this mode of protracted learning in the manufacturing of smaller-sized tools that does not necessarily require high level of physical strength as debated in several empirical studies regarding humans’ unique life history pattern (Bliege Bird & Bird, 2002; Jones & Marlowe, 2002).

Initiation ceremony is a period for intensive toolmaking skill training, along with other skills required for adulthood: Laguru, Mende, Eyak, Kaska, Pomo, Maleluka, Tukano, Goajiro, Ona. For example, Hamdani (Hamdani, 2001: 11-12) described the female initiation rites in Luguru, East Africa.

“The ‘mwali’ initiation rite... She explained that she literally lived in this hut for a period of three years... It was her mother who trained her to sort out green vegetables, to pound the maize and to prepare food. It was her aunts who taught her how to make sleeping mats, baskets, pottery and even caps. Mama Fulora explained that during her grandmother’s times, girls used to learn how to make clothing from barks of trees.”

Similarly, Hugh-Jones (1979: 86-87) provides a detailed narrative regarding the importance of initiation ceremony in toolmaking skill acquisition with the example Tukano people.

“For the initiates, the marginal period is a time of education... Then the elders systematically teach the initiates to make all the different kinds of basketry (carrying baskets, sieves, flat trays, cassava squeezers, etc.)... In addition to basketry, the initiates are taught how to make the feather head-dresses and other ritual ornaments used in dances... Just before the blowing of pepper, the initiates are taught to weave conical fish traps which they go and set in streams in the forest, accompanied by their masori.”

4.2 Task-dependent learning locations

While in most cases spatial contexts of toolmaking acquisition activities are underdocumented at best, there are sparse ethnographic narratives illustrating the spatial organization of complex skill acquisition like ground stone toolmaking and its relationship with natural resource distribution. For example, Cook (1968: 189-191) noted that different Zapotec villages have different ways of teaching the craft skill of metate and mano:

“In San Sebastian the finishing tasks in metate and mano production are taught in the residence lot; all preliminary and intermediate production tasks are taught and learned on location in the quarries. In San Juan, on the other hand, where the quarries are close to the village and more accessible than those in San Sebastian, metate size blocks of stone (trozos) are often carted from the quarry to the metatero’s residence lot in the village. Thus, the residence lot is, in many instances, the San Juan metatero’s workshop.”

This paragraph suggests that different toolmaking tasks are learned in distinct contexts, with raw material procurement typically occurring in quarries and subsequent processing tasks taking place in residential areas, depending on the distance between quarry and settlement. It suggests that evidence of learning and skill transmission may be spatially segmented, with early-stage production debris concentrated at resource extraction sites (e.g., Castañeda, 2018) and more refined production traces located in domestic settings (e.g., Knight, 2017). Archaeologists must therefore consider the spatial organization of artifact assemblages to identify stages of skill acquisition and social learning. Overlooking this spatial variability could lead to incomplete interpretations of how technical knowledge was transmitted.

4.3 Teaching and causal knowledge in the craftsmanship

Teaching and learning almost always appear together in modern education contexts. However, these two concepts are not exactly symmetrical in the sense that learning has a much broader connotation and is widely believed to present in an extensive range of species in the animal kingdom, while teaching is not. Or put it differently, learning does not necessarily require the presence of a teacher, while teaching, in its narrowest sense, implies the existence of a learner, even

if it is just an imaginative one. In fact, depending on the definition one adopts, it has been long debated that whether teaching is a cross-cultural or even cross-species phenomenon (Boyette & Hewlett, 2018). There have been several competing ideas around the presence and scope of teaching in human societies. The first position is represented by classical cultural and psychological anthropologists like Mead (1964) and Lancy (2010) who adopts a very narrow definition of teaching, arguing that teaching does not exist or is rare in small-scale cultures. Lancy (2010: 97) believed teacher is “nuanced, student-centered, developmentally appropriate instruction by dedicated adults” and is a recent product of “a long process of education changes”.

On the other hand, Hewlett and Roulette (2016) took a minimalist approach and defined teaching as “an individual modifies her/his behaviour to enhance learning in another”. Under this definition, teaching was identified through video recordings during as early as the infancy period in a variety of forms among the Aka hunter-gatherers of the Central African Republic and Republic of Congo, ranging from natural pedagogy to opportunity scaffolding to verbal instruction. Their research suggested that most teaching behaviors documented in the Aka foragers are usually very short and subtle, lasting only a few seconds but occurring repeatedly. The combination of short duration and observational subtlety of these types of teaching behaviors provide one possible explanation on why teaching was believed to be absent or rare for early anthropologists, in addition to the debates around the definition.

As observed in several non-industrial societies, including Kaska (Honigmann & Bennett, 1949: 185-186), Warao (Wilbert, 1976: 318-319), Kpelle (Lancy, 1975: 151), Haida (Blackman & Davidson, 1982: 182), and Tiv (Bohannon, 1958: 381), explicit transmission of causal knowledge appears to be rarely documented in non-industrial societies. Below is a direct quote from Honigmann (1949: 185-186), who provide positive evidence of the missing among Kaska foragers living in the arctic region.

“Education. Most education for living is received by the child in the course of performing routine activities. Such education is unformalized and undirected... Sometimes generalized instructions are furnished to children who, in carrying these out, often become confused. Thus we observed one fourteen year old girl make tea by putting the leaves into cold water. Nobody referred to the muddy quality of the beverage. The same kind of teaching was used with the ethnographer. Activities were illustrated but never explained in detail, the exposition of principles being ignored.”

These materials provide interesting insights into the longstanding debates about the nature and prevalence of teaching across cultures. If teaching is narrowly defined as the deliberate and structured transmission of explicit causal knowledge, which is essentially a Euro-centric definition, then its scarcity supports earlier claims by scholars like Mead (1964) and Lancy (2010) that teaching is very limited in small-scale societies. In these contexts, learning often occurs through observation, imitation, and participation rather than through formal instruction. As suggested by Hewlett and Roulette (2016), teaching in non-industrial societies may be subtle, brief, and embedded within daily activities, making it less visible to researchers employing narrow definitions. Therefore, recognizing the diversity of teaching practices requires adopting broader, more inclusive definitions that account for indirect and informal forms of knowledge transmission.

Additionally, these findings contribute to ongoing discussions in cultural evolutionary research, where experimental (Derex et al., 2019) and ethnographic studies (Harris et al., 2021) have suggested that explicit causal understanding is not necessary for the transmission of complex cultural traits (see Osiurak & Reynaud, 2020 for opposing views). It should be first clarified that causal knowledge here refers to exclusively the mechanical and physical knowledge regarding how materials interact with each other at the very basic level, which is different from causal perception that is prevalent in all kinds of indigenous ontologies. The recurring positive evidence of the lacking of causal knowledge transmission demonstrates in this project that individuals in non-industrial societies can effectively learn and transmit toolmaking skills without formal instruction in causal mechanisms, which means high-fidelity cultural transmission can occur through embodied practice and situated learning rather than explicit teaching. Recognizing that causal knowledge is not a prerequisite for cultural continuity challenges assumptions about the role of explicit reasoning in technological advancement and highlights the importance of contextual, experience-based learning in sustaining complex cultural traditions.

4.4 Alternative pedagogies: Co-production and miniature model

Instead of providing explicitly step-by-step causal knowledge-oriented teaching, a common way of acquiring a toolmaking skill is that experienced model co-produce an object with novice. When novice made mistakes, the experienced model take it back and correct the mistake and then return to the novice again for them to finish. During this process, no or little verbal explanation regarding why a specific step is necessary or how to avoid similar mistakes in future is

provided, as vividly described in Lancy's (1975: 151) childhood-focused dissertation fieldwork in Kpelle horticulturalists in Western Africa.

“Wollokolle's youngest wife, Goma was an expert at Bii paa or bag weaving... Now she is teaching a teenaged girl of 16 how to do it. The girl has prepared the strands of palm leaf and brought them to Goma to help her get started. The two alternate at weaving the bag. The girl makes a mistake and Goma unravels what she has done, does it correctly for awhile, then lets the girl take over again. Goma can't explain how to do it, either to me or the girl. She can only show and guide our hands as I, too, make an attempt to learn.”

On top of this Kpelle bag weaving case, similar social interaction dynamics has also been observed among Garo people in South Asia regarding how the skill of basket making is transmitted between male model and learners (Burling, 1963: 116).

“Only after he is about twelve years old does anything like formal teaching take place. The most difficult skill to learn or to teach is basket-making, which the father or some other man must impart to a boy. The instruction consists mostly of telling a child to make a particular basket as he has watched others do and then reprimanding him if he does it wrong. Occasionally a man will demonstrate, but I never saw anyone who was able to analyze and point out the mistakes another was making. If a person makes a mistake in a complex basket, for instance, his instructor will take the basket away, redo it himself, and then tell him to do it that way henceforth. No effort is made to break the complex weaving process down into steps so as to make it easy to understand and learn systematically, although it is recognized that certain baskets are easier to weave than others. Fathers tell their sons to make the easier types first, and then gradually lead them to the more difficult ones.”

This recurring phenomenon, as also documented in Shona (Aschwanden, 1982: 62), Tallensi (Fortes, 1938 : 60-61), Tzeltal (Nash, 1970: 274-275), Gros Ventre (Flannery, 1953: 159-160), and many other societies, has some deep implications for archaeologists who are interested in identifying the skill level of producers based on certain morphological traits of individual specimen or a whole artifact assemblage. In this pedagogical scenario, the experienced individual and

novice co-produce an item, and part or all of the mistakes made by novices can be erased by the former's behaviors. The coexistence of fossilized actions of both stakeholders preserved in archaeological materials often provide us a precious opportunity to reconstruct the complex learning behaviors in the past, be it stone artifacts (Assaf et al., 2016; Assaf, 2019) or pottery vessels (Crown, 2007).

Another commonly used pedagogical approach without resort to explicit causal knowledge transmission is the making of miniature models for and by children underage. To accommodate childrens' developmental constraint in hand size and physical strength, more experienced individual make smaller and often simplified version of the target artifact and then passed it to children as a model to begin with. It has been observed in many different societies and technological domains, including Blackfoot (Ewers, 1958: 103) bow and arrow making, Trobriands (Weiner, 1976: 192) net making, Jivaro (Harner, 1973: 92-93) pottery vessels and figurines, Mbuti carrying basket making (Turnbull, 1965: 125-126), etc. In terms of their implication for material culture research, these miniature models possess the interesting properties serving both as toys and tools (Lew-Levy et al., 2022; Riede et al., 2023), and their producers can be either adults or children.

4.5 Skill training for the community integration

Several sources in this study indicate that local communities perceive toolmaking not merely as a survival skill but as a vital process for social integration. Below are two short examples respectively from Tzeltal (Nash, 1970: 321) and Kpelle (Lancy, 1996: 154-155) illustrating toolmaking skill acquisition as a process of socialization.

“Learning agricultural tasks and pottery-making in the household is part of adjusting to membership in the community. Parents are responsible for teaching the value of industriousness and are held accountable throughout the life of the child. Agricultural tasks are scheduled in a yearly cycle that provides sufficient flexibility for a man to carry out community and familial ceremonial responsibilities.”

“In each of the previous cases, we see individuals acquiring weaving skills to enhance their own social participation. As Sua indicates, if a girl wants to “look nice,” to match

the appearance of women, she must learn the skills to dress and decorate herself. In attempting to do so, she will be accorded the status of a “legitimate peripheral participant.” She may be supplied with materials and someone competent may agree to model the skill and also to critique the fledgling product. Miniature tools may be supplied.”

This perspective fits into Lave and Wenger’s (1998) famous concept of “communities of practice,” where learning is inherently social and occurs through active participation in shared cultural practices. In these contexts, engaging in toolmaking allows individuals, particularly younger members, to become integrated into the social fabric of their community. Through observation, imitation, and guided practice, learners not only acquire technical skills but also internalize social norms, values, and identities associated with craftsmanship. This participation fosters a sense of belonging and reinforces communal bonds, demonstrating that toolmaking serves both practical and social functions. Thus, tool production becomes a medium through which individuals transition from peripheral participation to full membership within their cultural group, highlighting the dual role of skill transmission in sustaining both material needs and social cohesion.

5 Conclusion

Back to the six basic questions regarding toolmaking skill acquisition in non-industrial societies proposed in the introduction, here the author draw on key findings from both quantitative analyses and qualitative insights to provide brief answers as follows.

What: The two most frequent toolmaking skills are clothes making and basket making, documented in many non-industrial societies regarding of subsistence strategies or geographical distribution. From the perspective of raw material exploitation, prevalent yet perishable materials like plant fiber and wood are the dominant categories.

Who: People learn most toolmaking skills from their close kin members, particularly parents. Extended family members like grandparents and uncles/aunts are also heavily involved in this process.

When: For many skills, they started the learning processes in middle childhood (6-12 years old),

which can last for several years. The initiation ceremonies provide opportunities for a period of intensive training.

How: Although individual practice and trial and error always exist, toolmaking skill acquisition involves lots of social interactions, during which causal knowledge is rarely explicitly transmitted. Instead, two commonly used pedagogical approaches are 1) object co-production between experienced and inexperienced individuals as well as 2) miniature models manufactured for and by underage novices.

Where: Indoor learning scenarios are more frequently documented compared with the outdoor ones. The former is often associated with task involving meticulous motor control and specialized tools and materials, while activities demanding larger spaces or environmental resources are typically learned outdoors. For complex crafts involving multiple steps like raw material procurement, the locations are dependent on the specific task involved.

Why: This issue can be interpreted from two different angles, namely why people in non-industrial societies learn toolmaking skills and why people choose to learn them from certain individual. For the former question, toolmaking skill transmission is regarded as an important way of community integration on top of survival need. For the latter question, kinship is the main factor when choosing a model rather than expertise, prestige, or direct payoff comparison.

This project represents the most extensive compilation of its kind to date, offering unparalleled breadth and depth in documenting cultural transmission across diverse crafts and societies. Its comprehensive scope provides a valuable foundation for exploring patterns and dynamics in cultural learning and technological practices, enabling analyses that were previously unattainable due to limited data availability. This descriptive analysis also provides interesting insights into archaeological research of craft skill as well as some major theoretical issues in cross-cultural studies such as the scope of teaching in hunter-gatherer societies, the role of explicit causal knowledge in cultural evolution, community of practice, etc.

However, the presence of substantial and systematically patterned missing data introduces important limitations, which is not uncommon for cross-cultural dataset. The analysis indicates that missing values are not randomly distributed but are likely missing at random (MAR) or missing not at random (MNAR). This non-random missingness necessitates caution in conducting further statistical analyses, as it may bias results if not properly addressed. Careful application of imputation methods and sensitivity analyses will be essential to ensure the robustness and

reliability of future findings derived from this dataset. In the end, the author sees it as a stepping stone instead of a definitive text for a better understanding of the subject matter, which merely serves as a starting point for an open dialogue and a project to be collaborated with future researchers like the “Play Object Play” database (Riede et al., 2023).

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